

MATH 126 PROBLEM SET 2

Due: Sep 17, 11:59pm

1. Consider the initial value problem

$$\frac{d}{dt}u(t) = au(t), \quad u(0) = x_0 \tag{1}$$

- (1) Rewrite the equation as an integral equation. Recall the integral operator L used in Picard's iteration. Give the formula for L associated to (1).
(2) Let $u_0(t) = x_0$ and define $u_k = L^k u_0$. Show that

$$u_k(t) = x_0 \sum_{j=0}^k \frac{(at)^j}{j!}.$$

- (3) Solve (1) for t small.

2. Solve the following initial value problems and verify your solution.

- (1) $\partial_t u + \partial_x u = u^2$, $u(0, x) = h(x)$.
(2) $x\partial_y u + y\partial_x u = u$, $u(x, 0) = h(x)$.

3. Let $\mathbf{v} : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $w : \mathbb{R} \times \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}$. Suppose $u \in C^1(\mathbb{R} \times \mathbb{R}^n)$ satisfies

$$\frac{d}{dt}u(t, \mathbf{x}) + \mathbf{v} \cdot \nabla u(t, \mathbf{x}) + w(t, \mathbf{x}, u(t, \mathbf{x})) = 0.$$

The characteristics of this equation are given by

$$\frac{d\mathbf{x}}{dt}(t) = \mathbf{v}(t, \mathbf{x}(t)).$$

Show that

$$\frac{d}{dt}u(t, \mathbf{x}(t)) + w(t, \mathbf{x}(t), u(t, \mathbf{x}(t))) = 0.$$

Then use this to solve the equation

$$\partial_t u + x\partial_x + y\partial_y = u, \quad u(0, x, y) = h(x, y).$$

4. Exercises 2.3 and 2.4 in §2.7 of Borthwick.